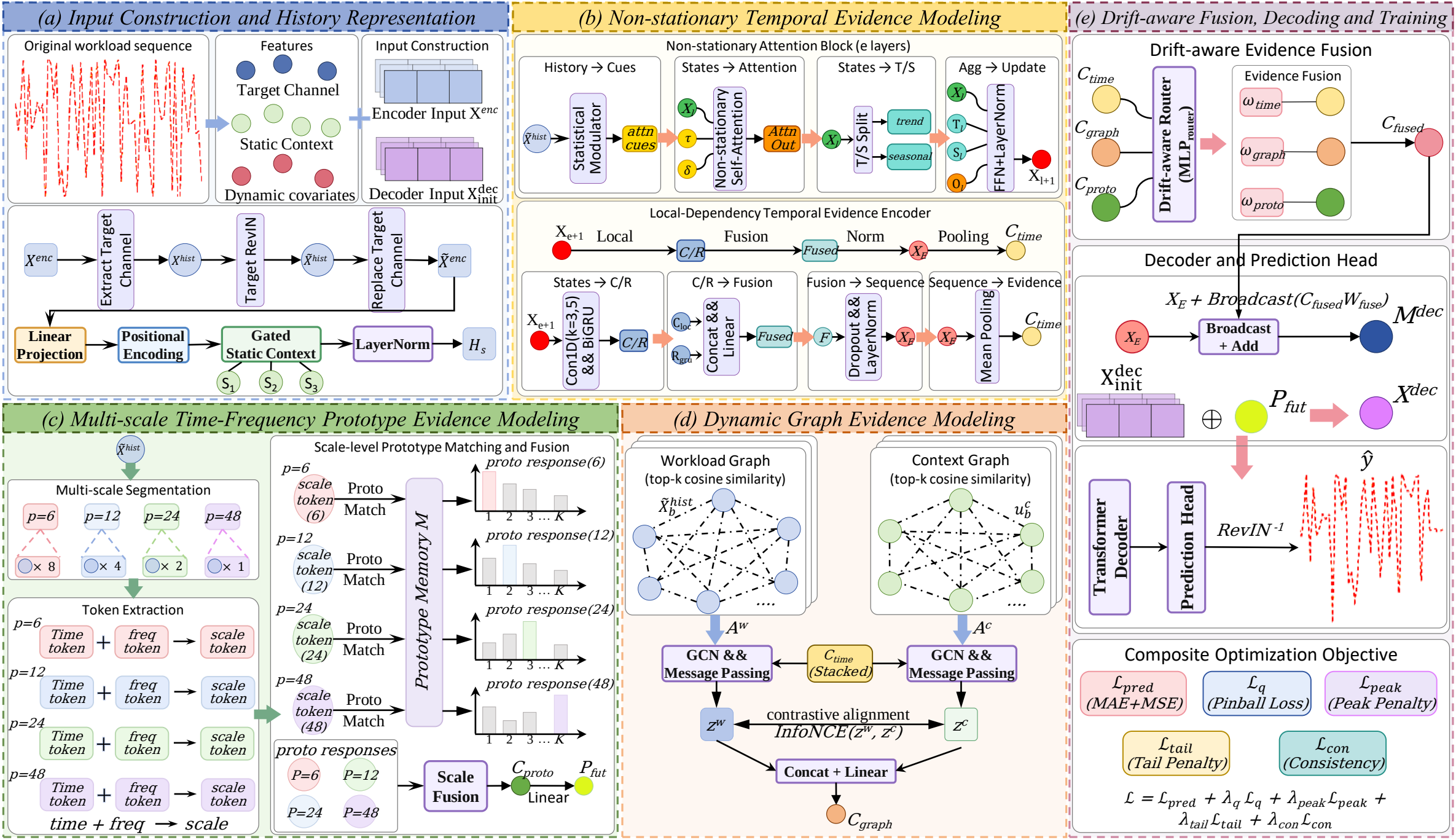

(a) Input Construction and History Representation
Original workload sequence
Features
Target Channel
Static Context
Dynamic covariates
Input Construction
Encoder Input X^enc
Decoder Input X^dec_init
X^enc
Extract Target Channel
X^hist
Target RevIN
X̃^hist
Replace Target Channel
X̃^enc
Linear Projection
Positional Encoding
Gated Static Context
S1
S2
S3
LayerNorm
H_s
(b) Non-stationary Temporal Evidence Modeling
Non-stationary Attention Block (e layers)
History → Cues
Statistical Modulator
attn cues
States → Attention
Non-stationary Self-Attention
Attn Out
States → T/S
T/S Split
trend
seasonal
Agg → Update
FFN+LayerNorm
X_{l+1}
Local-Dependency Temporal Evidence Encoder
X_{e+1}
Local
C/R
Fusion
Fused
Norm
X_F
Pooling
C_time
States → C/R
Con1D(k=3,5) && BiGRU
C/R → Fusion
Concat && Linear
Fusion → Sequence
Dropout && LayerNorm
Sequence → Evidence
Mean Pooling
(c) Multi-scale Time-Frequency Prototype Evidence Modeling
Multi-scale Segmentation
p=6
p=12
p=24
p=48
× 8
× 4
× 2
× 1
Token Extraction
Time token
freq token
scale token
time + freq → scale
Scale-level Prototype Matching and Fusion
scale token (6)
scale token (12)
scale token (24)
scale token (48)
Proto Match
Prototype Memory M
proto response(6)
proto response(12)
proto response(24)
proto response(48)
1 2 3 … K
proto responses
P=6
P=12
P=24
P=48
Scale Fusion
C_proto
Linear
P_fut
(d) Dynamic Graph Evidence Modeling
Workload Graph
(top-k cosine similarity)
Context Graph
(top-k cosine similarity)
A^w
A^c
GCN && Message Passing
C_time (Stacked)
GCN && Message Passing
z^w
contrastive alignment
InfoNCE(z^w, z^c)
z^c
Concat + Linear
C_graph
(e) Drift-aware Fusion, Decoding and Training
Drift-aware Evidence Fusion
C_time
C_graph
C_proto
Drift-aware Router (MLP_router)
Evidence Fusion
ω_time
ω_graph
ω_proto
C_fused
Decoder and Prediction Head
X_E + Broadcast(C_fused W_fuse)
X_E
Broadcast + Add
M^dec
X^dec_init
P_fut
X^dec
Transformer Decoder
Prediction Head
RevIN^-1
ŷ
Composite Optimization Objective
L_pred (MAE+MSE)
L_q (Pinball Loss)
L_peak (Peak Penalty)
L_tail (Tail Penalty)
L_con (Consistency)
L = L_pred + λ_q L_q + λ_peak L_peak + λ_tail L_tail + λ_con L_con